



Preliminary Ecological Appraisal

Serge Hill – Lunch Room and New Shed

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VERSION CONTROL

P01	29/07/2025	Keiron Young	Samuel Radonich	Approved with Comment	Minor Edits
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This report dated 16 September 2025 has been prepared for Studio Benjamin (the “Client”) in accordance with the terms and conditions of appointment (the “Appointment”) between the Client and Samuel Radonich (Greenlight Ecology Ltd) for the purposes specified in the Appointment. For avoidance of doubt, no other person(s) may use or rely upon this report or its contents, and Greenlight Ecology Ltd accepts no responsibility for any such use or reliance thereon by any other third party.



The site survey was undertaken by Samuel Radonich BSc (Hons) Consultant Ecologist.					
Date of survey	Temperature (°C)	Humidity (%)	Cloud Cover (%)	Wind (km/h)	Rain
21/07/2025	23	68	65	18	None

Ecological Survey Factor	Detailed using desk study and site survey (carried out under good weather conditions). Any specific limitations noted within relevant section. This table may include further work you will need to commission (if any) to obtain planning permission or comply with legislation for other consent. All clients are expected to read and understand this section, or to contact the lead surveyor for advice.
Conclusion, Impact or Recommendations	
Habitats and plants (see habitat map in appendix 1, location plan in appendix 2, proposal plan in appendix 3 and photos in appendix 4).	
Summary of Survey Findings (UKHab codes are used in the habitat descriptions)	Site Context The survey site is centred on National Grid Reference TL 10766 04254 and has an area of approximately 0.031ha (310 m²). The site comprises of an area of highly damaged, low quality modified grassland (dominated by perennial rye grass) which houses a mobile chicken hutch with gravel paths surrounding the meadows edge. A single storey barn with a pitched roof constructed from corrugated metal sheeting. A gravel path and a single immature wild cherry tree <i>prunus avium</i>. The site is situated within private land north of the Serge Hill Project in Bedmond, Abbots Langley. There are scattered trees to the east of the site and extending site edge at the border of the neighbouring arable field. Aerial imagery shows the local landscape to be primarily agricultural with pocket woodlands and woodland swathes in close proximity to the site. Water bodies can be found nearby in the form of a small swimming pond located west of the site ~270m and a man-made recreational swimming pool ~160m to the south. Such features will likely enhance the area for a variety of species, including bats, amphibians and reptiles. The site contains no features that are listed as a habitat of principal importance under Section 41 of the Natural Environment and Rural Communities (NERC) Act (2006) and no notable habitats are present within the site.



	No protected or notable plant species were recorded during the survey or during the desk study within 500m of the site within the last 5 years. No protected or notable species were recorded during the survey or during the desk study within 1km of the site within the last 5 years. On-site habitat descriptions <u>u1b – Artificial unvegetated; unsealed surface</u> A gravel trackway crosses the site and joins the existing roads (0.0013ha). <u>g4 – Modified grassland</u> There is a large section of modified grassland within the central location of the site located between the beech tree hedgerow and the woodland to the east (0.010ha). The grassland is heavily managed and in poor condition with large bare patches created using a mobile chicken hutch and dominated with perennial rye grass <i>Lolium perenne</i>. <u>u1b5 -Developed land; Sealed surface – Buildings</u> A single-story barn with a 45-degree pitched roof. Constructed from corrugated metal sheeting. Corrugated roofing sheets were laid directly onto the wooden roofing timbers, with no roofing membrane or insulation (0.008ha). <u>g4 – Modified grassland~ Rural Tree</u> A single immature wild cherry tree <i>prunus avium</i> situated within the area modified grassland.
<i>Foreseen Impacts</i>	On-site habitats The habitats on-site are varied and not notable, however the proposed development will result in the loss of ~0.010ha vegetated habitat. This is not likely to have a significant impact on biodiversity considering the habitat being lost and the surrounding area. As there are extensive trees within the locality, the habitat loss is unlikely to significantly impact biodiversity or alter the migratory routes used by varied species as they use this site as passing through the local area. Notable habitats No impacts to any notable habitats are anticipated due to the small scale and distance of the proposed development from such habitats.
<i>Recommendations</i>	On-site habitats Retained (where possible) trees should be protected in line with the measures outlined in the British Standard "Trees in Relation to Design, Demolition and Construction to Construction - Recommendations" (BS 5837) (2012).



	A biodiversity net gain (BNG) report has been produced for the proposal, as more than 25m² of habitat is affected by the proposal (if less than 25m² then the proposal could be exempt) (Greenlight Ecology, 2025). Notable habitats Best practice measures to minimise the possibility of pollution affecting the nearby ancient woodland must be implemented during construction. A Construction Environment Management Plan (CEMP) may be required for this.
Locality and Designated Sites	
<i>Summary of Survey Findings</i>	On-site designations The site is not subject to any designation. Statutory designated sites (within 2km) There are no known statutory sites located within 2km of the site Non-statutory designated sites There are multiple known Non-statutory sites within 2km of the site and are listed as follows; <u>Ancient Woodland Inventory Sites.</u> Appspend Wood – TL 112 053, Area – 4.91ha, Distance – 1080m Blackwater Wood – TL 094 057, Area – 3.24ha, Distance – 1900m Hanging/Piercornerwoods – TL 105 034, Area – 2.02ha, Distance – 720m Hanging/Piercornerwoods – TL 106 035, Area – 1.06ha, Distance – 640m Hanging/Piercornerwoods – TL 104 037, Area – 12.30ha, Distance – 420m Holt Wood – TL 117 035, Area – 1.47ha, Distance – 1050m Madams/Birch Woods – TL 119 055, Area – 8.24ha, Distance – 1580m Madams/Birch Woods – TL 118 057, Area – 13.48ha, Distance – 1740m Park Wood – TL 126 053, Area – 13.12ha, Distance – 1960m Scrubs Wood – TL 123 050, Area – 2.09ha, Distance – 1720m <u>Herts and Middlesex Wildlife Trust Nature Reserve.</u> File Code – 75/013x, Name – Longspring Wood, Area – 1.24ha, Distance – 1910m <u>Local Wildlife Sites.</u> File Code – 66/007, Name – Blackwater Wood, Area – 3.24ha, Distance – 1900m



	File Code – 67/002, Name – Potters Crouch Plantation, Area – 53.98ha, Distance – 370m File Code – 67/004, Name – Prae Wood, Area – 110.40ha, Distance – 1930m File Code – 67/020, Name – Park Wood, Area – 19.75ha, Distance – 1900m File Code – 67/021, Name – Scrubs Wood, Area – 2.09ha, Distance – 1720m File Code – 67/022, Name – Long Spring, Area – 1.00ha, Distance – 1710m File Code – 67/024, Name – Appspound Wood, Area – 4.91ha, Distance – 1080m File Code – 67/025, Name – Birch Wood, Area – 21.57ha, Distance – 1580m File Code – 75/013, Name – Long Spring Wood, Area – 1.22ha, Distance – 1910m File Code – 75/014, Name – Bedmond Green, Area – 3.30ha, Distance – 1190m File Code – 75/036, Name – Great Notley, Area – 2.06ha, Distance – 1310m File Code – 75/061, Name – Moor Wood, Area – 1.37ha, Distance – 1470m File Code – 76/006, Name – Wellfield Spring, Area – 0.84ha, Distance – 250m File Code – 76/007, Name – Job’s Wood, Area – 3.17ha, Distance – 640m File Code – 76/010, Name – Potterscrouch Section, Area – 0.53ha, Distance – 850m File Code – 76/011, Name – Holt Wood, Area – 1.49ha, Distance – 1050m File Code – 76/012, Name – Winch Hill Wood, Area – 2.41ha, Distance – 1420m File Code – 76/016, Name – Piecorner and Hanging Wood, Area – 13.27ha, Distance – 420m File Code – 76/033, Name – Featherbed Lane Copse, Area – 0.88ha, Distance – 30m File Code – 76/084, Name – Tenements Farm Area, Area – 0.00ha, Distance – 1050m File Code – 76/114/01, Name – Serge Hill Meadow, Area – 1.48ha, Distance – 70m <u>Regionally Important Geological / Geomorphological Site (RIGS).</u> File Code – 76/010, Name – Pottercrouch Section, Area, 0.18ha, Distance – 840m
<i>Foreseen Impacts</i>	On-site designations No impacts foreseen. Statutory and non-statutory designated sites No impacts to non-designated sites are anticipated due to the small scale and distance of the proposed development from such sites (where known) as well as the rural location of the site with surrounding physical barriers e.g – roads and arable landscape.
<i>Recommendations</i>	On-site designations None required. Statutory and non-statutory designated sites None required.



Invasive / Non-native species	
Summary of Survey Findings	There were no non-native species under Schedule 9 of the Wildlife and Countryside Act 1981 located within the site during the survey or during the desk survey.
Foreseen Impacts	Construction will not result in the spread of non-native / invasive species under Schedule 9 of the Wildlife and Countryside Act 1981.
Recommendations	Continue to monitor and survey for any non-native / invasive species under Schedule 9 of the Wildlife and Countryside Act 1981 prior to construction.
Invertebrates	
Summary of Survey Findings	The habitats present on-site, including modified grasslands and individual tree likely provide limited foraging and shelter opportunities for common invertebrates.
Foreseen Impacts	The grassland habitat mentioned above will be removed during construction. The loss of such habitats could result in a reduction in invertebrate habitat and could result in the fragmentation of the local landscape.
Recommendations	No further surveys. Suggested biodiversity enhancements The incorporation of bee bricks (e.g. Ibstock BeeHabitat or similar alternative brand) into the fabric of the new buildings would provide sheltering opportunities for pollinators. These should be installed 0.5m above ground level on a south-facing elevation with no obscuring vegetation. The site could be further enhanced via the provision of native wildflowers or wildflower turf, which would provide foraging opportunities for invertebrates.
Bats	
Summary of Survey Findings	EPSL data A search of the magic.gov.uk database for granted EPSLs within a 2km radius of the site has been completed alongside an intensive desk study and record centre analysis. No EPSLs are present within a 2km radius of the site, and no records of bat activity have been documented in the last 5 years. Foraging and commuting habitat



	Habitats recorded immediately adjacent site are assessed to provide foraging and commuting opportunities for bats in the form of native hedgerow, agricultural land surrounding the site, lines of semi-mature and mature trees. These habitats are likely to provide micro-climatic conditions that support invertebrates that will in turn provide foraging opportunities for local bat populations. Most notably, the trees (both mature and immature) on site adding to the continuity of vegetated linear features present in the wider landscape. Bats are well known to utilise linear features to aid navigation whilst travelling between foraging resources and roost sites. Roosting habitat The single-story corrugated metal barn was subject to a internal and external Preliminary Roost Appraisal (PRA) by James Sweetman (NE Class license number 2021-51575-CLS-CLS) and was assessed as having low potential to support roosting bats. In line with the BCT guidelines the barn a single emergence survey was carried out on 21st July 2025, no emergences were recorded (BCT, 2023). No evidence of roosting bats was identified on or within the individual tree on site or in the surrounding area. Please refer to separate bat report for full details of PRA and emergence survey (Greenlight Ecology, 2025).
<i>Foreseen Impacts</i>	Roosting habitat (construction plans) The proposed development will result in the construction of a new shed and change of use of an existing barn. The existing barn that is to be refurbished has been assessed as having low potential to support individual bats therefore there is a low risk of roosting bats being present when the works commence. Roosting habitat [Trees] No features were identified on any of these trees adjacent to the site and as such there are unlikely to be any impact to bats. Foraging and commuting habitat The proposed development will result in the loss of small areas of suboptimal foraging habitat in the form of modified grassland but given the low value and the presence of more extensive areas of foraging and commuting habitat in the locality, this is likely to be inconsequential for bats. Artificial lighting The proposed development may lead to an increase in the amount of current lighting of surrounding habitats or the retained building without mitigation. This may disturb commuting bats but is deemed minimal.
<i>Recommendations</i>	



	Roosting habitat A Preliminary Roost Assessment of the existing barn was advised and was carried out on 21st July 2025, the barn was assessed as having low potential to support roosting bats and was subject to one emergence survey which recorded no emergences, please see separate bat report for full details (Greenlight Ecology Ltd, 2025). Advisory of no bat emergence/re-entry surveys are required on mature trees within the adjacent woodland during the active bat season (May – September) to confirm presence/likely-absence of bats. This is due to the trees being classified as negligible for roosting bats In the unlikely event that a bat or evidence of bats is discovered during the development all work must stop and a bat licensed ecologist contacted for further advice. Foraging and commuting habitat No further surveys are required. Artificial lighting A low impact lighting strategy is advised to be adopted for the site during post-development which outlines the areas of the site that will be retained as dark corridors. Parameters can be found on the Bat Conservation Trust website: Suggested biodiversity enhancements The installation of three bat boxes at the site will provide additional roosting habitat for bats. The bat boxes will be incorporated into the fabric of the new development or the surrounding area. Suitable bat boxes include Habibat Bat Box, Ibstock Enclosed Bat Box or similar alternative brand. Bat boxes should be positioned 3-5m above ground level facing in a south or south-westerly direction with a clear flight path to and from the entrance, away from artificial light.
Birds	
<i>Summary of Survey Findings</i>	Trees and Vegetation No evidence of active nesting birds was identified on site and habitats recorded on site are assessed to provide limited nesting opportunities for breeding birds. Barn owls The site does not provide any suitable nesting sites for barn owls.



	Overwintering birds Due to the small size of the site and the extent and type of the habitats recorded, the site not considered suitable to support a significant assemblage of protected and/or notable birds.
<i>Foreseen Impacts</i>	trees The proposed development could result in the destruction or the disturbance and subsequent abandonment of active bird nests. However, no active nests were noted during the survey and works are due to be carried out prior to the nesting bird season. Barn owls None foreseen. Overwintering birds None foreseen.
<i>Recommendations</i>	trees Any building or vegetation removal should be undertaken outside the period 1st March to 31st August. If this timeframe cannot be avoided, a close inspection of the vegetation should be undertaken immediately, by a qualified ecologist, prior to the commencement of work. All active nests will need to be retained until the young have fledged. Precautions should be taken with machinery and noise levels when working close to any retained nests so as not to disturb any nearby nesting birds during construction works. At least a 3-5m buffer should be created between any machinery and active nests until the young have fledged. Barn owls Owing to the nature of the proposed development and the low potential for impacts to barn owl, further surveys are considered to be disproportionate. Overwintering birds None required. Suggested biodiversity enhancements None required



Reptiles	
Summary of Survey Findings	EPSL data A review of the MAGIC database returned no granted EPSL records for protected reptiles within 2km of the site. Habitat suitability Habitats recorded on site are assessed to provide no suitable foraging, commuting, and refuge opportunities for reptiles. It is noted that due to recent management of grassland on site which retained a short sward length at the time of survey, the grassland is currently suboptimal for reptiles due to an absence of refuge opportunities. The site is also enclosed by intensely managed agricultural land in addition to hardstanding tracks. These landscape features are likely to significantly limit dispersal opportunities onto the site for reptiles from the wider landscape. However, although the majority of enclosing agricultural fields are subject to intense management including grazing and regular ploughing, the fields are likely to retain narrow unmanaged field margins interconnected by hedgerows that may provide foraging and commuting opportunities for a limited number of reptiles. Although potential connectivity to the site for reptiles is limited in extent, the presence of a small population of common reptiles cannot be fully discounted, albeit limited to peripheral vegetation.
Foreseen Impacts	Although a small area of sub optimal habitat is being removed as part of the development, there is a low risk that a low number of reptiles could be present in the vicinity of the works. These could be injured or killed without mitigation and thus care must be undertaken during construction and overseen a suitably qualified ecologist.
Recommendations	None required. A precautionary working method is recommended to be implemented for widespread reptiles during construction, including the following measures: • Vegetation will be maintained at a short sward (5cm) to discourage reptiles. • Any excavations will be covered overnight, or a ramp will be installed to enable any trapped animals to escape. • Best practice pollution prevention measures will be implemented to minimise impacts to nearby habitats. • Any chemicals or pollutants used or created by the development should be stored and disposed of correctly according to COSHH regulations. • If any reptiles are found in the working area these should be allowed to disperse of their own accord or, if at immediate risk, should be moved by hand to a sheltered, vegetated area away from disturbance. • In the unlikely event that a reptile is identified, works must cease and advice must be sought from a suitably qualified ecologist. Suggested biodiversity enhancements The site could be enhanced for reptiles post-development with the inclusion of log piles (created from felled materials) and planting of areas of native shrubs, to provide sheltering opportunities.



Amphibians	
<i>Summary of Survey Findings</i>	EPSL and survey data A review of the MAGIC database returned no granted EPSL records for great crested newts within 2km of the site, however desk study analysis returned records of surveys carried out 1320m from site resulting in positive identifications. Further, no positive class survey licence return or DLL historic survey data (2017 – 2022) were present within 2km of the site. Habitat suitability A swimming pond is located approximately 270 m to the west of the site. This pond was surveyed; however, it is regularly used as a swimming pond and is surrounded by extensive suitable terrestrial habitat for amphibians. Given the high-quality habitat in the immediate surrounding area and the limited connective habitat between and suitable habitat within the site's redline boundary, the pond has not been considered further in the assessment. Additionally, there is a man-made recreational pool located 160 m to the south of the site, but this feature does not provide any characteristics suitable for amphibians. While common amphibians may still be present within areas of suitable habitat on-site due to their ability to travel significant distances from breeding ponds, only limited opportunities for foraging and sheltering exist, primarily in areas of planting and dense vegetation. No hibernation features were identified within the redline boundary.
<i>Foreseen Impacts</i>	Given the lack of suitably connected breeding ponds within 500m of the site, the presence of GCN on-site is considered unlikely and therefore impacts to amphibians as a result of the proposed development are deemed to be acceptably low.
<i>Recommendations</i>	None required. Suggested biodiversity enhancements The site could be enhanced for amphibians post-development through creation of amphibian hibernacula using rubble and logs from site clearance. Information on how to construct a hibernaculum can be found here: https://www.wiltshirewildlife.org/hibernaculum
Badger	
<i>Summary of Survey Findings</i>	No badger setts were noted on site or within a 30m radius of the site. The site is considered unsuitable for badgers given the lack of suitable sett excavation areas/ground. Further, there is limited suitable badger foraging habitat on site despite the occurrence of fruiting trees/scrub. The site is also surrounding by arable farmland, which is likely potential habitat therefore increasing the likelihood of badgers being present within the surrounding area of the site.
<i>Foreseen Impacts</i>	No impacts are anticipated on badgers as a result of the proposed development.



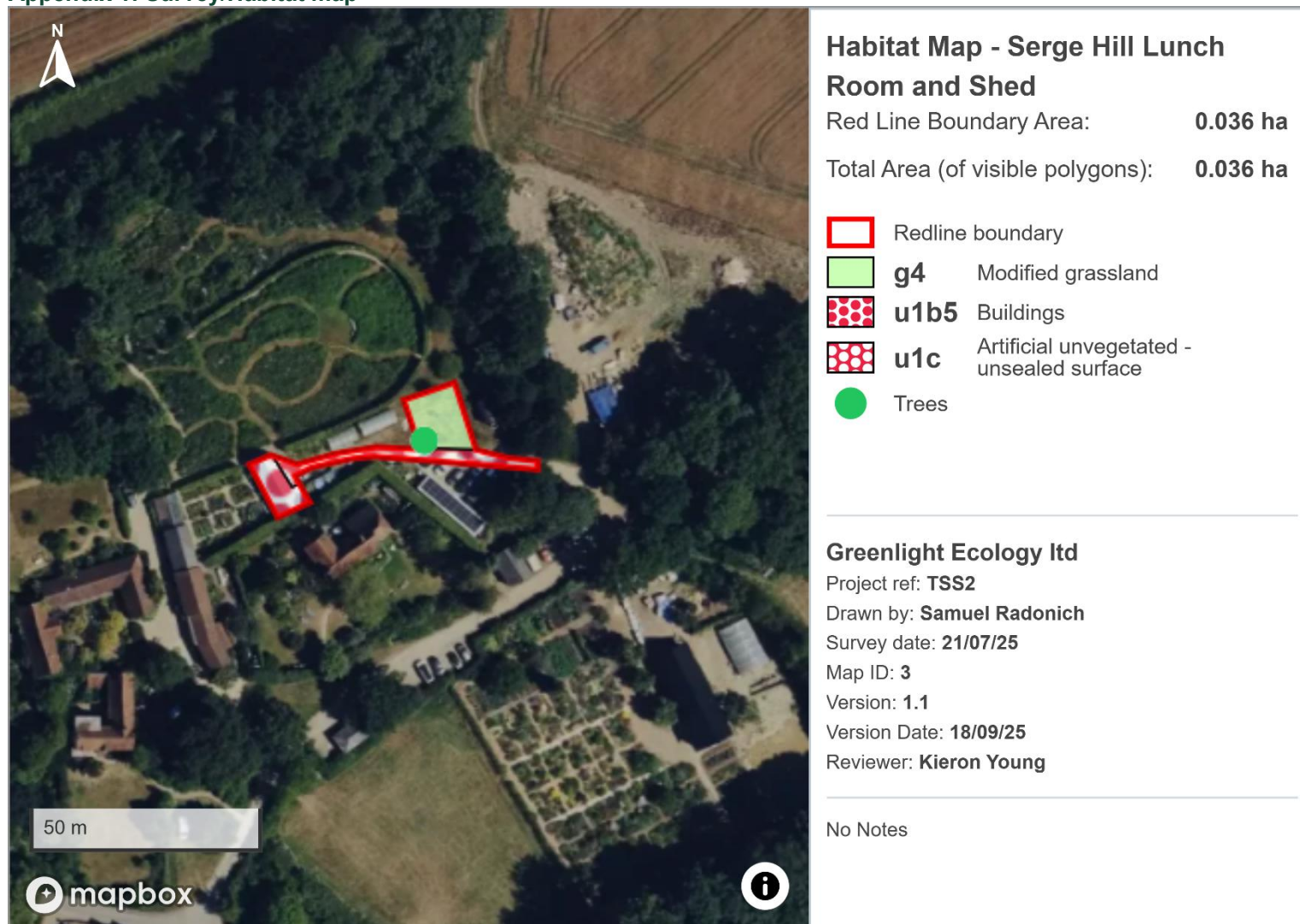
<i>Recommendations</i>	None required, however, owing to the nature of badger ecology and their mobile nature. the proposed development and the low potential for impacts to badgers, further badger surveys are considered to be disproportionate. A precautionary working method will be implemented during construction, including the following measures: • A toolbox talk will be given to contractors regarding the possible presence of badgers at the site. • A pre-commencement inspection of the site will be undertaken for any new badger activity if works do not commence within three months. • Heras fencing will be erected around the working area to prevent encroachment into retained habitats where badger setts could be present. • Any excavations will be covered overnight, or a ramp will be installed to enable any trapped animals to escape. • The use of night-time lighting will be avoided, or sensitive lighting design will be implemented to avoid light spill on to retained habitats which badgers could use. • Any chemicals or pollutants used or created by the development should be stored and disposed of correctly according to COSHH regulations. In the unlikely event that a badger sett is identified, works must cease and advice must be sought from a suitably qualified ecologist. Suggested biodiversity enhancements Planting more fruit bearing trees and species-rich grassland to increase foraging opportunities for badgers.
Riparian animals	
<i>Summary of Survey Findings</i>	A review of the MAGIC database returned no granted EPSL records for otters or water voles within 2km of the site. There are no water courses on or connected to the site. There are also no riparian habitats present on site or within an influencing distance.
<i>Foreseen Impacts</i>	No impacts are anticipated on riparian animals as a result of the proposed development.
<i>Recommendations</i>	None required. Suggested biodiversity enhancements No suggestions as there is no watercourse in close proximity to the development.
Hazel dormouse	
<i>Summary of Survey Findings</i>	EPSL data A review of the MAGIC database and desk study analysis returned no granted EPSL records for hazel dormice within 2km of the site.



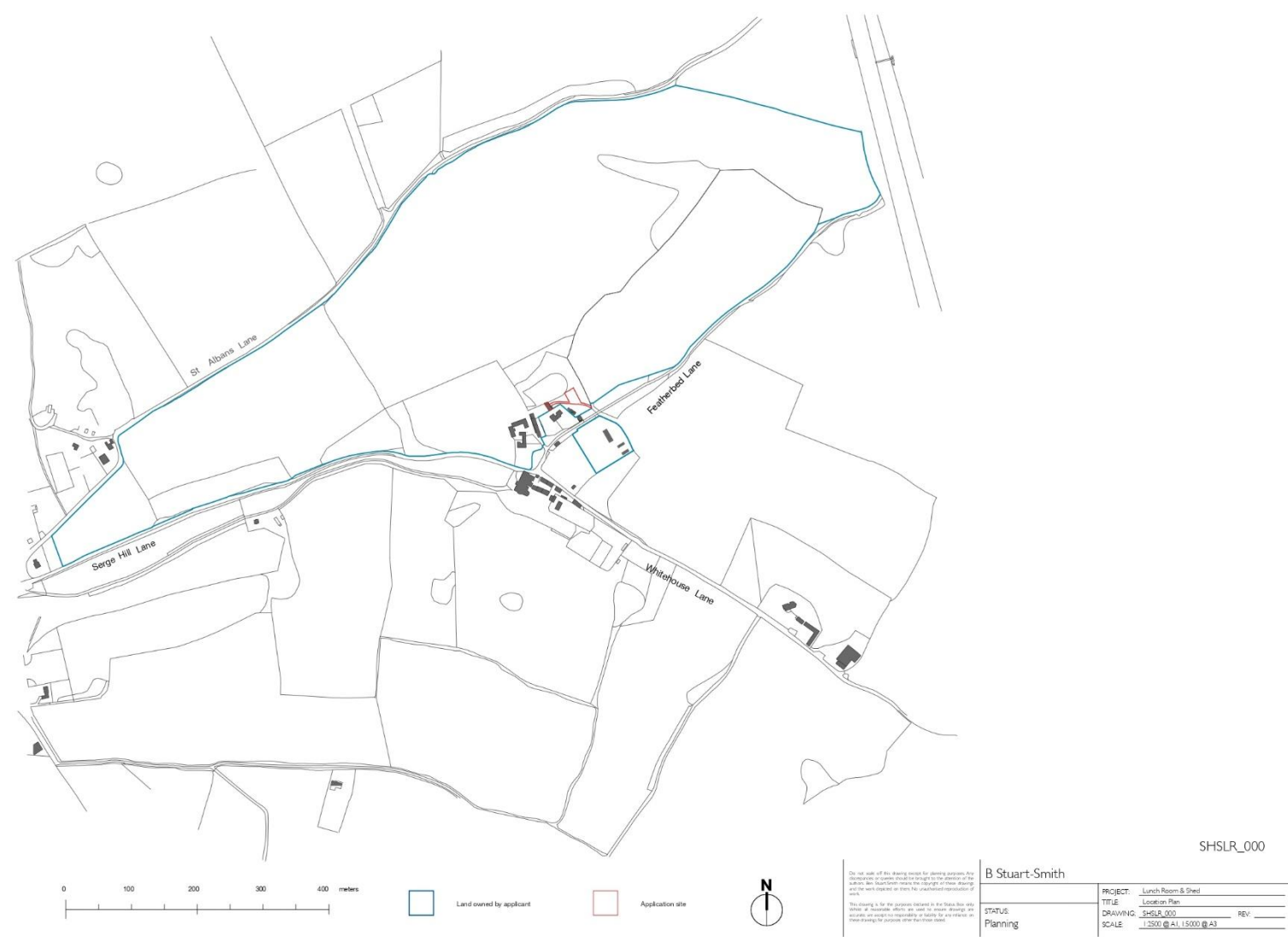
	Habitat suitability The site lies within of the known current range for hazel dormice, however there are no suitable habitats within the development area. The surrounding habitat of pocket deciduous woodlands and arable farmlands have no connectivity to the site with the exception of a roadside hedge line. As such it is considered likely that hazel dormice are absent from site.
<i>Foreseen Impacts</i>	No impacts are anticipated on hazel dormice as a result of the proposed development.
<i>Recommendations</i>	None foreseen. In the unlikely event that a dormouse or evidence of dormouse is identified, works must cease and advice must be sought from a suitably qualified ecologist.
Other e.g. hedgehog	
<i>Summary of Survey Findings</i>	The surrounding agricultural lands of the site have the potential for foraging and commuting opportunities for hedgehogs, with woodland habitat nearby.
<i>Foreseen Impacts</i>	Grassland will be removed during construction. The loss of such habitat is likely to be inconsequential to local hedgehog populations owing to their low value and the presence of more extensive habitat locally. However, construction activities could result in the death or injury of hedgehogs, if present.
<i>Recommendations</i>	Similar to the badgers, a precautionary working method will be implemented during construction, including the following measures: - Any excavations will be covered overnight, or a ramp will be installed to enable any trapped animals to escape. - The use of night-time lighting will be avoided, or sensitive lighting design will be implemented to avoid light spill on to retained habitats which hedgehogs could use. - Any chemicals or pollutants used or created by the development should be stored and disposed of correctly according to COSHH regulations. If any hedgehogs are found in the working area these should be allowed to disperse of their own accord or, if at immediate risk, should be moved by hand to a sheltered, vegetated area away from disturbance. Suggested biodiversity enhancements The following habitat creation and enhancement opportunities could be incorporated into the proposed development which would be beneficial for hedgehogs: - Planting fruit bearing trees and species-rich grassland to increase foraging opportunities. - Creation of brash piles or installation of hedgehog houses in shady areas. - Installation of gaps under boundary fencing to enable hedgehogs to move freely through the site.



Appendix 1: Survey/Habitat map



Appendix 2: Location map



Appendix 3: Proposed plan



Appendix 4: Photos

Photograph	Description
	Figure 1: Showing Mixed Woodland (adjacent to site boundary so surveyed but not mapped)



**Figure 2:
Showing
Mixed
Woodland
(adjacent
to site
boundary
so
surveyed
but not
mapped)**



Figure 3:
Showing
Mixed
Woodland
(adjacent
to site
boundary
so
surveyed
but not
mapped)



Figure 4:
Showing
Mixed
Woodland
(adjacent
to site
boundary
so
surveyed
but not
mapped)



Figure 5:
Showing
Mixed
Woodland
(adjacent
to site
boundary
so
surveyed
but not
mapped)



**Figure 6:
Showing
Mixed
Woodland
(adjacent
to site
boundary
so
surveyed
but not
mapped)**



**Figure 7:
Showing
Mixed
Woodland
(adjacent
to site
boundary
so
surveyed
but not
mapped)**



Figure 8:
Showing
Mixed
Woodland
(adjacent
to site
boundary
so
surveyed
but not
mapped)



**Figure 9:
Showing
Mixed
Woodland
(adjacent
to site
boundary
so
surveyed
but not
mapped)**

	Figure 10: Showing Modified Grassland.



Figure 11:
artificial
unvegetate
d; unsealed
surface



Figure 12:
Showing
Existing
Barn
Eastern
aspect



Figure 13:
Showing
existing
barn
northern
aspect



Figure 14:
Showing
existing
barn
southern
aspect



Figure 14:
Showing
existing
barn
western
aspect

